IRINA COJOCARI

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EDUCATION

Instituto Superior Técnico — Lisbon, Portugal
BSc in Computer Science and Engineering | 2025 – Present (Expected 2028)
- **Coursework:** Operating Systems, Analysis and Synthesis of Algorithms, Object-Oriented Programming, Data Structures and Algorithms, Computer Architecture, Logic for Programming.

TECHNICAL SKILLS

- **Languages:** C, C++, Python, Java, Assembly, Prolog
- **Systems & Tools:** Arch Linux (Daily Driver), NixOS, Void, Git, Neovim, Linux CLI, Makefile
- **Spoken Languages:** Portuguese (Native), English (Fluent)

PROJECTS

RISC-V Transformer Simulation | *RISC-V Assembly, Logisim* | [Link]
- Implemented the forward pass of a simplified Transformer (Self-Attention) entirely from scratch in RISC-V Assembly.
- Built matrix multiplication and dot product routines manually, including strict integer overflow detection.
- Utilized system calls for file I/O to parse text files into memory buffers for embedding lookups.

CloudIST Virtual Machine Simulator | *C, POSIX, Linux*
- Developing a concurrent VM simulator in C that manages resource allocation across multiple physical servers.
- Utilizing POSIX file descriptors for file I/O and implementing multi-process execution (`fork`, `exec`, `setrlimit`) alongside multi-threading for parallel processing.

Express Retail Platform (ERP) | *Java, UML*
- Architecting a large-scale retail management system in Java, applying strict OOP principles (Open-Closed) for high extensibility.
- Designing comprehensive UML class diagrams for the core and application packages.
- Implementing Java Serialization for state persistence, custom exceptions, and polymorphic behavior for dynamic pricing.

Tower Building Algorithm | *C++*
- Developing an optimized algorithm to calculate the maximum height of a 3D tower with strict stability and rotation constraints.
- Implementing an iterative dynamic programming approach to prevent stack overflow and optimize time/space complexity.

Discord Announcement Bot | *Python, Webhooks* | [Link]
- Developed a custom Discord bot to scrape and forward university announcements, saving server resources via webhooks.
- Designed the tool to scale for multiple courses and IST communities, using `.env` files for secure configuration.

Invoice Handler | *C, Makefile* | [Link]
- Built a robust invoice management system in C to parse, store, and manipulate records efficiently.
- Separated interface and implementation using modular header files (`.h`) and Makefiles.

Scrabble Game | *Python* | [Link]
- Developed a terminal-based Scrabble game with multi-language support (English, French, Portuguese) and optimized the game loop.

Logic Puzzle & Scheduler | *Prolog* | [Link]
- Developed algorithms in Prolog for analyzing data and generating schedules based on specific constraints.

Flocking Simulation | *Python* | [Link]
- Created a graphical simulation emulating bird flight patterns and flocking behavior with custom visual assets.

Automated CV Pipeline | *Python, Typst, GitHub Actions* | [Link]
- Built a CI/CD pipeline that automatically generates a PDF resume from a structured `data.json` file.
- Wrote a Python script to parse JSON data and inject it into a Typst template, completely separating content from formatting.
- Configured GitHub Actions to automatically compile and commit the updated PDF on every `git push`.

ACTIVITIES & EXPERIENCE

Security Team Técnico (STT) | *Member* | 2026 – Present
- Participate in weekly cybersecurity workshops and compete in Capture The Flag (CTF) challenges.
- Apply practical skills in cryptography, web exploitation, and reverse engineering.

IEEE-IST Computer Society | *Member* | 2026 – Present
- Collaborating with the web development team to maintain and update the organization's website.

Ser Cientista Research Program | *Universidade de Lisboa* | 2025 | [Certificate]
- Collaborated with a PhD student on the Al4Sci project, applying Python to Artificial Intelligence research in Sciences.